EE69210: Machine Learning for Signal Processing Laboratory
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# Kernel Density Estimation

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**Keywords:** Gaussian Mixture Models

**Grading Rubric**

| | Tick the best applicable per row | | | Points |
|---|---|---|---|---|
| | Below Expectations | Lacking in Some | Meets all Expectations | |
| Completeness of the report | | | | |
| Organization of the report (5 pts) | | | | |
| Quality of figures (5 pts) | | | | |
| Creation of Dataset (10 pts) | | | | |
| Histogram plots and density mappings. (25 pts) | | | | |
| Plot showing how changing the bins can morph the modality of data from unimodal to bimodal, etc. (20 pts) | | | | |
| Visual representation of the data distribution and the density around each point in the data. (25 pts) | | | | |
| Smoothened KDE. (10 pts) | | | | |
| | | | **TOTAL (100 pts)** | |

# 1. Kernel Density Estimation

Kernel Density Estimation (KDE) is a *non-parametric* technique used to estimate the probability density function (PDF) of a random variable based on a finite set of samples. Unlike parametric methods, KDE does not assume any predefined form of the distribution and instead constructs the estimate directly from the data. Let the observed dataset be denoted as:

$$\mathbf{X} = [x_1, x_2, \ldots, x_n]^\top \in \mathbb{R}^{n\times 1}, \tag{1}$$

where each $x_i \in \mathbb{R}$ is a scalar data point and $n$ is the number of samples.

Our objective is to estimate the underlying probability density function $p(x)$ of the distribution that generated these samples. The kernel density estimator at a query point $x \in \mathbb{R}$ is given by:

$$\hat{p}_h(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right), \tag{2}$$

where:

- $h > 0$ is the **bandwidth** (smoothing) parameter,
- $K(\cdot)$ is the **kernel function**, typically a symmetric PDF,
- $\hat{p}_h(x)$ is the estimated density at point $x$.

## 1.1 Gaussian Kernel

A widely used kernel is the Gaussian kernel, defined as:

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right). \tag{3}$$

Using this kernel in the KDE formula results in:

$$\hat{p}_h(x) = \frac{1}{nh\sqrt{2\pi}}\sum_{i=1}^{n} \exp\left(-\frac{(x - x_i)^2}{2h^2}\right). \tag{4}$$

## 1.2 Vectorized Formulation

To express the computation in vectorized form, we define a distance tensor between the query point $x$ and the dataset:

$$\mathbf{D}_x = x\mathbf{1} - \mathbf{X} \in \mathbb{R}^{n\times 1}, \tag{5}$$

where $\mathbf{1} \in \mathbb{R}^{n\times 1}$ is a column vector of ones.

The kernel values for all data points can then be computed simultaneously:

$$\hat{p}_h(x) = \frac{1}{nh\sqrt{2\pi}} \mathbf{1}^\top \exp\left(-\frac{\mathbf{D}_x \odot \mathbf{D}_x}{2h^2}\right), \tag{6}$$

where $\odot$ denotes element-wise multiplication and the exponential is also applied element-wise.

## 1.3 Bandwidth Selection

The choice of bandwidth $h$ critically affects the smoothness of the density estimate. A small $h$ may result in overfitting (high variance), while a large $h$ may cause underfitting (high bias). A common heuristic for bandwidth selection is Silverman's rule of thumb:

$$h = 1.06\,\hat{\sigma}\,n^{-1/5}, \tag{7}$$

where $\hat{\sigma}$ is the empirical standard deviation of the dataset.

# 2. Data Preparation and Visualization

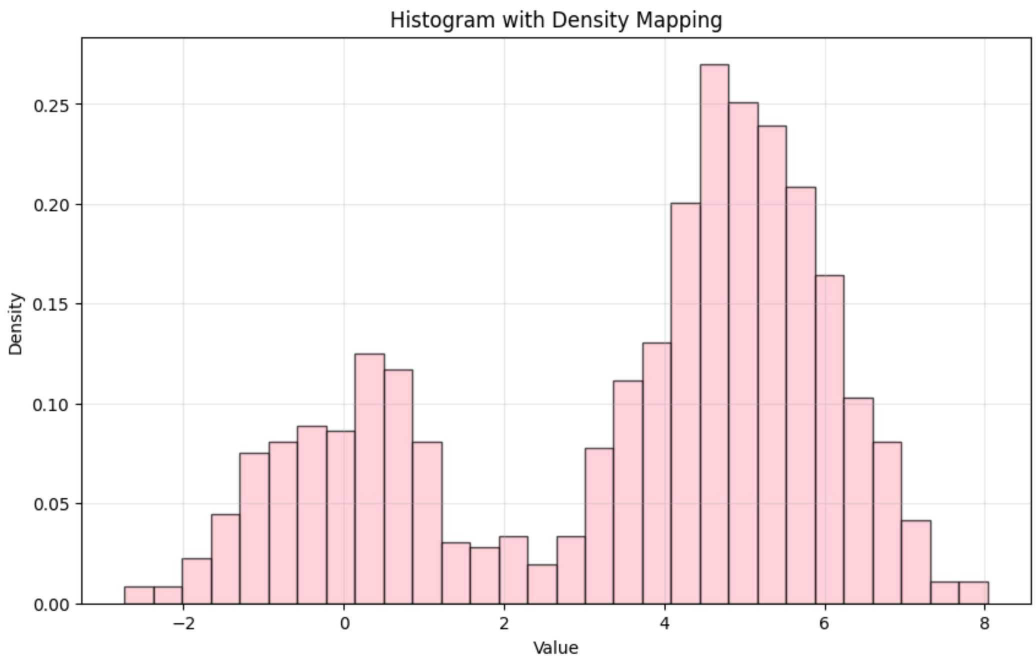


Fig. 1. Density histogram of 1000 data points showing bimodal distribution.

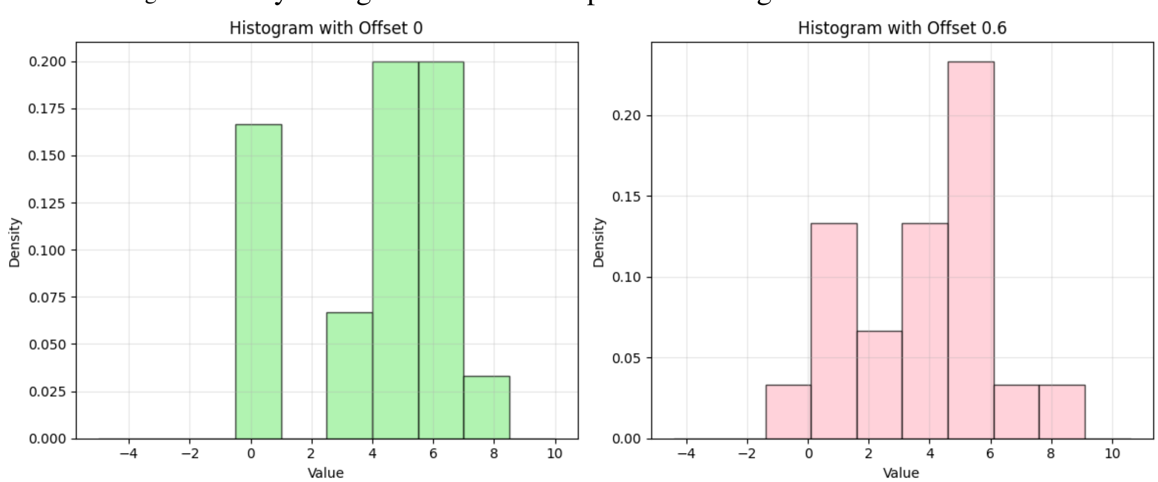


Fig. 2. Comparison of histogram densities with different bin offsets (0 vs 0.6) on a small dataset of 20 points

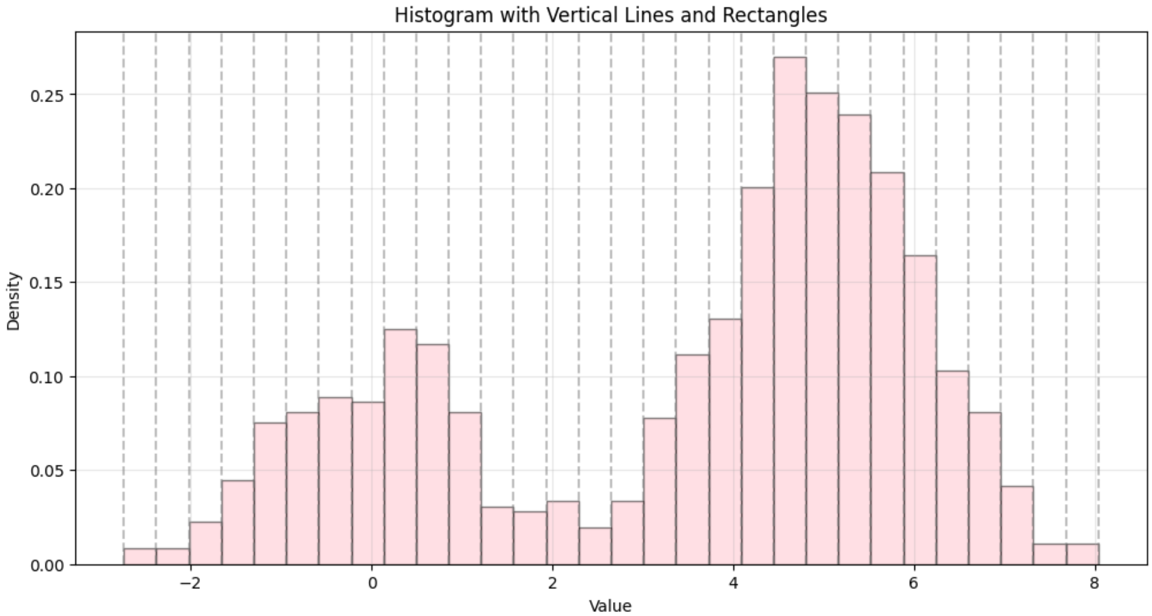


Fig. 3. Histogram represented with vertical lines at bin edges and rectangles to visualize both the bin boundaries and the density distribution.

## 3. Kernel Density Estimate Plot: Custom Implementation vs SciPy

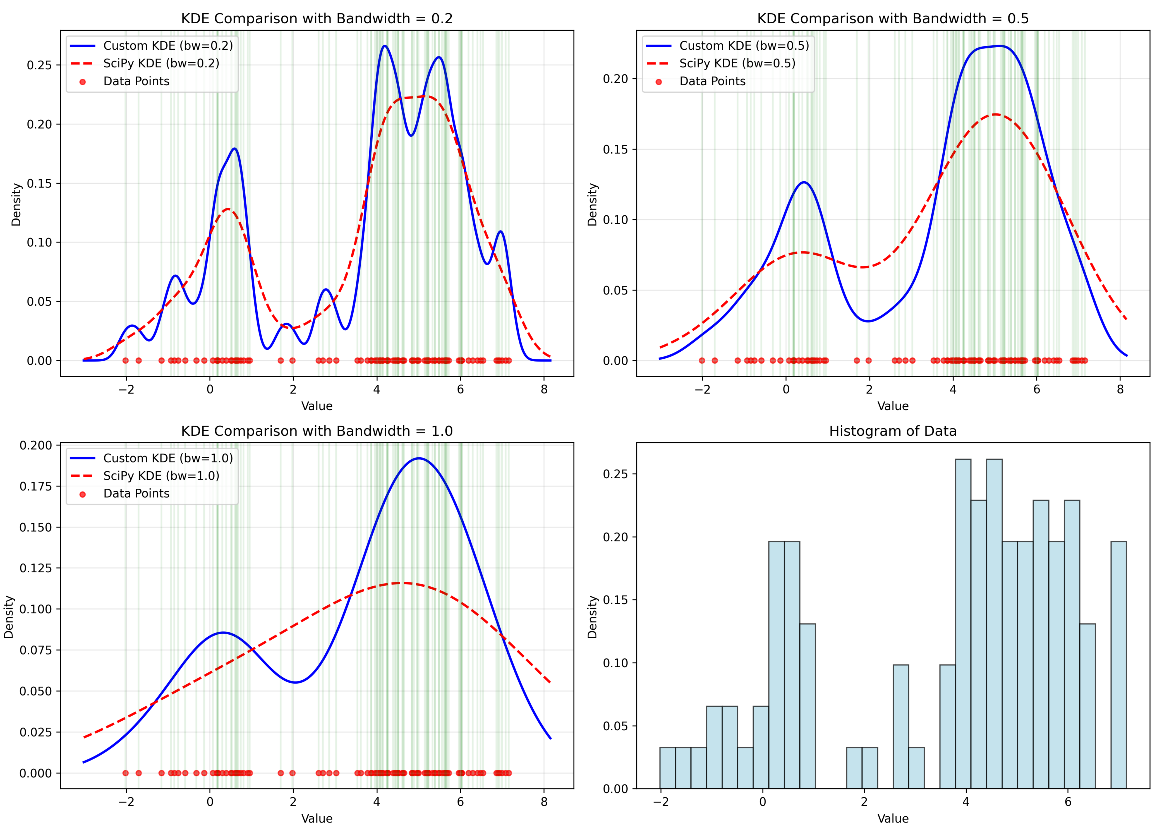


Fig. 4. Plot representing the KDE of a given data sequence for different bandwidths. The plots show comparison of custom and SciPy implementation